YAT COLLEGE

YAT College · 175 Cremorne St, Cremorne VIC 3121

Solution Design

LMS Replacement — GrayBoard to DOODLE (on-premises)

Engagement	YAT LMS Replacement Project — Phase 1
Document type	Technical design (Solution Design)
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Implemented by	the LMS Replacement Deployment Report
Classification	Commercial-in-confidence

Contents

1. Purpose and Scope

This document specifies the solution for replacing YAT's end-of-life GrayBoard LMS with the DOODLE platform, deployed on-premises at the Cremorne campus. It translates the board-approved recommendation from the LMS Replacement Business Case into an implementable design covering the infrastructure, the DOODLE installation approach, the migration of student records from GrayBoard, and the cutover.

In scope of this design

- On-premises infrastructure for DOODLE at the Cremorne campus (server, storage, network placement, backup).
- DOODLE installation and base configuration; integration with Active Directory, Office 365 and AVETMISS reporting.
- Migration of all student records from the legacy GrayBoard LMS, with reconciliation.
- The cutover plan, scheduled into the December–January teaching break.

Out of scope of this design

- Cloud hosting — noted as a future consideration, not part of this engagement.
- Application-layer high availability / clustering.
- Ongoing operational support after acceptance (separately contracted).

2. Design Inputs and Requirements

2.1 Inputs

- LMS Replacement Requirements — the functional and non-functional targets.
- Engagement Role Brief — scope, phases, on-premises deployment direction.
- ICT Environment Overview — the current campus environment the LMS sits within.

2.2 Requirements the design must meet

Requirement	Target / note
Replacement platform	A current, vendor-supported LMS — DOODLE (selected via market evaluation)
Deployment	On-premises at the Cremorne campus; Windows Server 2016 hosting standard
Record preservation	All GrayBoard student records migrated with full integrity (counts reconcile)
Availability	≥ 99% (a meaningful step up from the degrading GrayBoard)
Accessibility	WCAG 2.1 Level AA (Disability Discrimination Act 1992)
Integrations	Active Directory SSO · Office 365 SMTP · AVETMISS/ASQA export · LDAP user groups
Performance	Match or exceed GrayBoard, including ~3× concurrent load at end-of-term peaks
Vendor support	Severity-1 response ≤ 4 h (teaching hours) / ≤ 8 h (outside)
Cutover	During the December–January teaching break; training delivered before cutover

3. Review of Existing Architecture

Not applicable — this is a replacement, not a hardening of an existing system — GrayBoard is decommissioned, not extended. The assessment of the legacy GrayBoard platform is in the LMS Replacement Business Case.

4. Architecture Design

4.1 Assumptions and constraints

#	Assumption / constraint	Source
A1	Deployed on-premises at the Cremorne campus (current operating model)	Role Brief
A2	Windows Server 2016 application-hosting standard	Replacement Requirements
A3	All GrayBoard student records must be preserved with integrity	Standards for RTOs 2015
A4	Authentication via Active Directory; preserve LDAP group structures	Replacement Requirements
A5	Cutover confined to the December–January teaching break	Replacement Requirements

4.2 Deployment environment

The DOODLE server is deployed in the YAT Cremorne campus computer room — physically secured, air-conditioned, UPS-protected — consistent with the existing ICT environment. It sits in the staff network zone, with student-zone access to the LMS application governed by Active Directory permissions.

4.3 Identity and access

- DOODLE authenticates against Active Directory via LDAP bind — single sign-on for staff and students.
- Existing LDAP user-group structures are preserved (per the User Access Policy in force).
- MFA for staff with grading or course-management roles, per the User Access Policy.

4.4 Network

- The server uses the redundant campus network (dual NICs) with no single point of failure at the network layer.
- Students reach the LMS across the zone boundary under AD permissions; staff reach it from the staff zone.
- Office 365 SMTP relay for LMS notifications; outbound access for vendor updates via the campus firewall.

4.5 Compute (server)

- A new mid-range enterprise server (dual PSU, RAID-protected disks), running Windows Server 2016, hosts DOODLE and its MySQL database.
- Sized to match or exceed GrayBoard's performance, including the ~3× concurrent load during end-of-term assessment windows.

4.6 Load balancing

Not applicable — single-server on-premises deployment — no load balancer in scope (this is not a horizontally-scaled or highly-available design).

4.7 Database

- MySQL on the application server (DOODLE's supported database); engine version confirmed against DOODLE's compatibility matrix.
- Populated by migrating the student records from the legacy GrayBoard LMS (see §5), with post-migration reconciliation.

4.8 Storage

- RAID-protected local disk for the OS, the DOODLE application, and the MySQL database.
- The campus NAS (RAID-5) holds course attachments, student submissions, and backup copies.

4.9 Security

- Active Directory permissions and the two-zone separation control access; antivirus/EDR on the server.
- HTTPS for LMS application traffic; TLS for the MySQL connection.
- WCAG 2.1 Level AA conformance at the application layer (a DOODLE selection criterion).

4.10 Monitoring

The existing YAT system-management and monitoring server tracks the DOODLE server's health and runs the nightly backups; availability is tracked against the ≥ 99% target.

4.11 Naming and asset conventions

The server and related assets are recorded in the YAT ICT asset register under the standard naming and ownership conventions.

4.12 Backup

- Nightly backup of the DOODLE database and application data to the system-management server, with offsite tape rotation — aligned to the existing Backup and Recovery Process.
- GrayBoard is retained (read-only) until acceptance, as the migration rollback position (see §5).

4.13 Recovery objectives

Recovery is backup-based: nightly backups give an RPO of up to one day, with restore from the system-management server or offsite tape. The design targets ≥ 99% availability — a step up from the degrading GrayBoard.

4.14 Components requiring vertical scaling

The server is provisioned with headroom for growth; storage on the NAS is expandable. A future capacity increase is a server resource upgrade in a maintenance window.

4.15 Single points of failure removed

Not applicable — this is a supported single-server replacement, not a high-availability design; resilience beyond the existing campus posture (redundant network, RAID, UPS) is out of scope — cloud HA is noted as a future consideration.

4.16 Configuration decisions left to the implementer

#	Decision	Why left open
C1	Server specification (CPU / RAM / disk)	Size against the GrayBoard performance baseline + peak load
C2	MySQL engine version	Confirm against the DOODLE compatibility matrix
C3	Storage sizing (local + NAS)	Compute from the migrated data footprint + growth

C4	Backup schedule specifics	Align with the existing Backup and Recovery Process windows

5. Implementation Sequencing

The replacement is a change to a live service — GrayBoard is in use until cutover — so the sequence and rollback matter. The work is confined to the December–January teaching break, with GrayBoard retained read-only as the rollback position until acceptance.

#	Step	Impact	Verification	Rollback
1	Provision and harden the server; install Windows Server 2016	None (parallel)	Server build checklist	n/a
2	Install and configure DOODLE; wire AD / O365 / AVETMISS integrations	None (parallel)	Integration tests	n/a
3	Migrate GrayBoard student records; reconcile counts + sample fidelity	None (parallel)	Record reconciliation	Re-run migration
4	Parallel-run DOODLE alongside GrayBoard; deliver staff training	None	Acceptance testing; trained staff	Continue on GrayBoard
5	Cutover to DOODLE (Dec–Jan break); redirect users	Planned downtime in break	Smoke + acceptance	Revert to GrayBoard
6	Decommission GrayBoard after acceptance	None	Sign-off received	n/a

6. Simulation and Verification Plan

Not applicable — no high-availability failure/resize simulation applies to this single-server on-premises design; verification (record reconciliation, performance baseline, accessibility audit) is documented in the Deployment Report's testing section.

7. Out of Scope

- Cloud hosting — a future consideration, not this engagement.
- Application-layer high availability / clustering.
- Ongoing operational support after acceptance of the Closure Pack (separately contracted).

8. References

- LMS Replacement Requirements; Engagement Role Brief; ICT Environment Overview.
- Privacy / Data Handling Policy; User Access Policy; Backup and Recovery Process.
- Standards for RTOs 2015 (record retention); Disability Discrimination Act 1992 (WCAG 2.1 AA).

Document control

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Successor document	LMS Replacement Deployment Report (Phase 2–3)